M01 - A02

Wednesday, 17 June 2026 8:41 AM

Engineering Memo A02
Intelligence Revision: Variable Corridor Costs
Cost Model Update — Issued Following Amendment A01

Classification: RESTRICTED — Autonomous Systems Squad (ASS) only

Issued by: Emberlight Rescue Authority, Engineering Division

Subject: Non-uniform traversal cost model — multi-wing facility

Situation Update

Detailed structural analysis of the Emberlight facility has revealed that corridor traversal costs are **not uniform** across the facility. The three wings were constructed under different conditions and use different materials, resulting in distinct cost profiles for each sector.

CRUDY-1's propulsion system confirms the following per-wing cost models:

Wing	Cost model	Description
Wing Alpha	Uniform: $w = 1$	Standard construction throughout. No change from Memo 01.
Wing Beta	Depth-based: $w = 1 + \lfloor c_{\text{max}}/3 \rfloor$	Reinforced inner sectors. Cost increases as CRUDY-1 moves deeper into the wing. c_{max} is the higher column index of the two connected cells (columns 0–9, left to right).
Wing Gamma	Seed-randomised: $w \in \{1, 2, 3, 4, 5\}$	Emergency construction — irregular materials. Each corridor carries a fixed but seed-determined weight. Your Marimo workbook displays the weights for your facility.
Inter-wing	Fixed: $w = 1$	Access corridors between wings retain uniform cost.

The **total traversal cost** of a route is the sum of all corridor weights along the path. CRUDY-1's mission directive requires it to find a route to an exit that **minimises total energy expenditure**, not simply the number of corridors traversed.

Engineering Advisory — Algorithm Adequacy

Your current traversal algorithm (designed in Memo 01) operates under the assumption that all corridors have equal cost. Under that assumption, minimising the number of steps is equivalent to minimising total cost, and breadth-first search is sufficient.

This equivalence **no longer holds**. A route with fewer corridors may cost more than a longer route through cheaper sectors. Any algorithm that does not account for edge weights when selecting the next node to explore **cannot guarantee an optimal solution**.

You must select and justify an algorithm that is capable of finding minimum-cost paths in a weighted graph. Your justification must explain why your chosen algorithm is correct for this problem, not merely that it is a known weighted-path algorithm.

weighted-path algorithm.

Required Actions

Complete each revision in your Amendment A02 Marimo workbook. Your original A01 responses remain visible for reference.

[A2-1] — Revise Data Model for Weighted Edges Criterion 3 · 150–300 words

Revise your data model to represent non-uniform corridor costs. Your response must:

- Identify which ADT definitions from your A01 model need to change to support weighted edges.
- Specify how edge weights are stored and retrieved. Is the weight a property of the edge, a separate lookup structure, or computed on demand?
- Explain how the per-wing cost model is represented. Does your data model treat Wing Beta and Wing Gamma differently at the ADT level, or does it store weights uniformly and compute them at construction time?
- Justify your design choice.

[A2-2] — Revise Algorithm for Weighted Traversal Criterion 4 · 150–300 words

Select and justify a traversal algorithm capable of finding minimum-cost paths in your weighted multi-wing facility. Your response must:

- Name and briefly describe your chosen algorithm.
- Explain why this algorithm is *correct* for the problem — i.e. why it guarantees a minimum-cost path given the cost models above.
- Identify the key difference between your Memo 01 algorithm and this revised algorithm. What structural change makes the difference?
- Discuss one limitation or trade-off of your chosen algorithm in this specific context.

[A2-3] — Revised Pseudocode Criterion 4

Write revised pseudocode for your weighted traversal algorithm. Your pseudocode must clearly show:

- how edge weights are retrieved during traversal,
- the priority or ordering mechanism that distinguishes this algorithm from BFS,
- how the algorithm terminates and returns the optimal route.

[A2-3b] — Revised Implementation Criterion 4 · runnable code

Implement your revised algorithm in the Marimo code cell provided. Run it on your facility and display the minimum-cost route and its total cost. The output must show:

Implement your revised algorithm in the Marimo code cell provided. Run it on your facility and display the minimum-cost route and its total cost. The output must show:

- the sequence of cells in the optimal route,
- the cumulative cost at each wing transition,
- the total route cost.

Required Actions

[A2-4] — Evaluate Solution Quality Criterion 5 · 100–200 words

Briefly evaluate the quality of your revised solution:

- Is your algorithm guaranteed to find the *globally* optimal path across all wings, given that each wing uses a different cost model?
- How does the depth-based cost in Wing Beta affect the likelihood that CRUDY-1 traverses it directly versus routing through other wings?
- Identify one scenario (based on your seed) where the minimum-cost route is *not* the shortest-hop route, and explain why.

Submission

Complete all four actions in your **Amendment A02 Marimo workbook** and save responses to `responses_A02.json`. Submit at **Observation 5**.

Note: Your original Memo 01 and Amendment A01 responses are visible in the A02 workbook for reference. You do not need to repeat them — only address what changes.

The facility costs more to navigate than it looks.